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PAPER

nf-core/viralmetagenome: A Novel Pipeline for Untargeted Viral Genome Reconstruction

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Abstract

Motivation: ~~Eukaryotic viruses present significant challenges for genome reconstruction and variant analysis~~ Reconstructing eukaryotic viral genomes from metagenomic data is computationally challenging due to their ~~extensive~~ high genetic diversity and potential ~~genome~~ for segmentation. ~~While de novo assembly followed by~~ Current approaches often rely on labor-intensive manual curation for reference ~~database matching~~ selection and scaffolding~~is a commonly used approach~~, ~~the manual execution of this workflow is extremely time-consuming, particularly due to the extensive reference curation required~~limiting scalability for large studies or rapid outbreak response. ~~Here, we address the~~ There is a critical need for ~~an~~ automated, scalable ~~pipeline~~ workflows that can efficiently ~~handle~~ reconstruct viral ~~metagenomic analysis~~ genomes without ~~manual intervention~~prior knowledge of the target.

Results: We present nf-core/viralmetagenome, a comprehensive ~~viral metagenomic~~ Nextflow pipeline for the untargeted ~~genome~~ reconstruction and variant analysis of eukaryotic DNA and RNA viruses. ~~Viralmetagenome is implemented as a Nextflow workflow that processes short-read metagenomic samples to automatically detect and assemble viral genomes, while also performing variant analysis.~~ The pipeline ~~features~~ automates the entire process from read preprocessing to consensus generation, integrating multiple *de novo* assemblers, automated reference selection, and iterative consensus refinement. It features robust quality control~~metrics~~, ~~comprehensive~~ extensive documentation, and seamless ~~integration with containerization technologies, including~~ portability via Docker and Singularity. We ~~demonstrate~~ validated the ~~utility and accuracy of our approach through validation~~ pipeline on ~~both~~ diverse simulated and real datasets, ~~showing robust performance across diverse viral families in~~ demonstrating its ability to recover high-quality genomes from complex metagenomic samples and resolve co-infections, making it a powerful tool for viral discovery and surveillance.

Availability: nf-core/viralmetagenome is freely available at https://github.com/nf-core/viralmetagenome with comprehensive documentation at https://nf-co.re/viralmetagenome

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Supplementary information: Supplementary data are available at https://github.com/Joon-Klaps/nf-core-viralmetagenome-manuscript online.

Key words: viralmetagenome, bioinformatic pipeline, nextflow, viral metagenomics, viral assembly, viral variant analysis

Introduction

Reconstructing viral genomes from metagenomic sequencing data presents considerable computational challenges, particularly for viruses exhibiting extensive genetic diversity (Baaijens et al. 2017, Meleshko et al. 2021). This ~~diversity is further compounded by segmented genomes in families like~~ challenge is amplified in segmented viruses such as influenza, rotavirus, and bunyaviruses, where individual segments can evolve under distinct selective pressures and ~~reassort, contributing to a complex landscape for genome~~ reassortment complicates reconstruction. While ~~pipelines are often designed to~~ some pipelines target specific viruses and their subtypes (Shepard et al. 2016), accurate ~~and complete~~ genome reconstruction of samples with unknown references typically requires time-consuming manual curation of contigs and reference matching (de Vries et al. 2021). This ~~manual curation process is time-consuming, making it impractical for~~ makes large-scale metagenome studies or rapid ~~response scenarios that involve emerging viral outbreaks of unknown origin~~outbreak responses impractical.

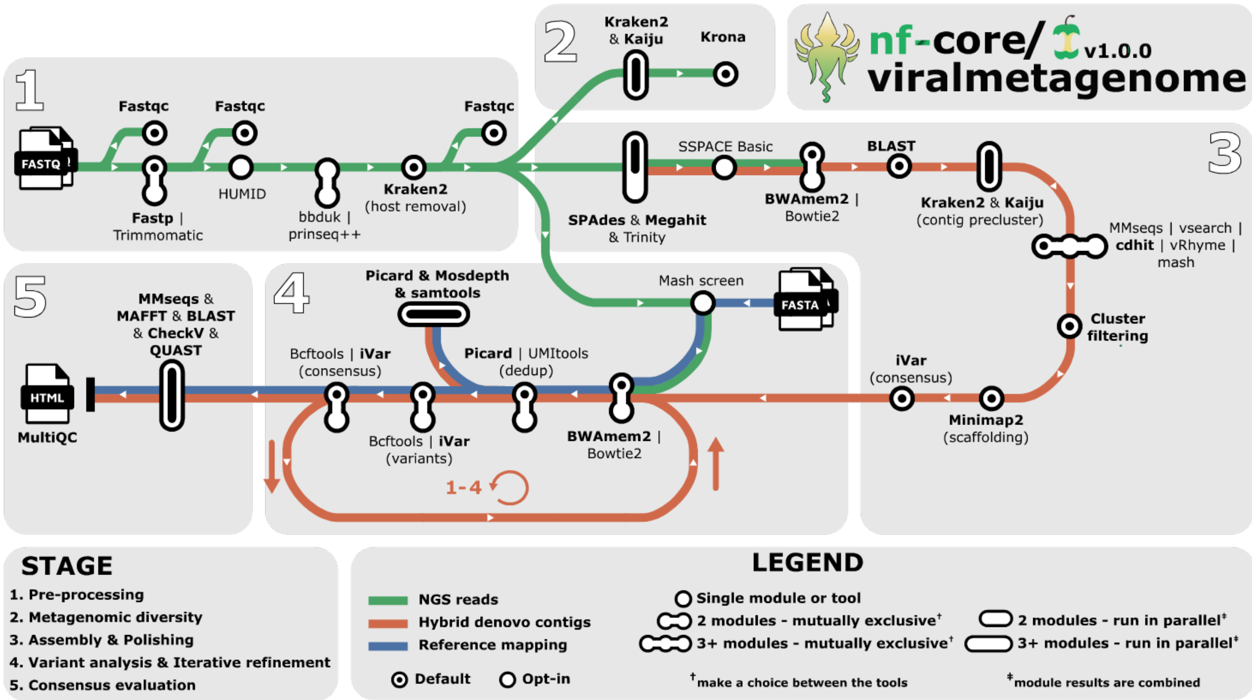

Fig. 1. ~~Visual overview of the~~ **Overview of the nf-core/viralmetagenome pipeline for untargeted viral genome reconstruction.** nf-core/viralmetagenome ~~pipeline for untargeted viral genome reconstruction. nf-core/viralmetagenome~~ processes short-read data through pre-processing, metagenomic diversity assessment, ~~*de novo*~~ de novo assembly with multiple assemblers, scaffolding with automated reference identification and ~~contig~~ taxonomy-guided clustering of contigs, and iterative consensus refinement through read mapping and variant calling. Quality control metrics, assembly statistics, and coverage data are integrated into interactive MultiQC reports and standardised overview tables for downstream analysis~~.~~

To address these limitations, we developed nf-core/viralmetagenome, a comprehensive pipeline specifically designed for untargeted viral genome reconstruction. The pipeline is developed using Nextflow (Di Tommaso et al. 2017) within the nf-core framework ~~(Ewels et al. 2020)~~(Ewels et al. 2020, Langer et al. 2025), ensuring reproducibility through containerization with Docker (Merkel 2014) and Singularity (Kurtzer et al. 2017), and enabling portability across computational platforms such as local desktops, high-performance clusters and cloud environments.

Pipeline Description

nf-core/viralmetagenome implements an automated workflow ~~performing~~ for *de novo* assembly, reference matching, and iterative consensus refinement ~~for the reconstruction of~~ to reconstruct viral genomes without prior target knowledge. The pipeline consists of five major stages: read preprocessing, metagenomic diversity assessment, contig assembly and scaffolding, iterative consensus refinement with variant analysis, and quality control (Figure 1). ~~While this manuscript highlights key differences between particular tools, unless~~ Unless otherwise specified, the pipeline offers multiple options to accommodate established user workflows and preferences. ~~In depth tool details are available~~ Detailed tool descriptions are provided in Supplementary Table 1.

Read preprocessing

~~Input reads provided via~~ Input reads listed in sample sheets containing sample names and short-read FASTQ paths are preprocessed using FastQC and ~~have their adapters trimmed with~~ trimmed for adapters using fastp (Chen et al. 2018) (default) or Trimmomatic (Bolger et al. 2014). Fastp ~~is overall faster and has~~ offers automated adapter detection and ~~trimming~~ faster processing (Chen et al. 2018). UMI deduplication is implemented using HUMID (Laros) ~~and once readsare mapped to a reference with~~ for raw reads, or UMI-tools after mapping (Smith et al. 2017). Multiple sequencing runs ~~are~~ can be merged after trimming by specifying merge group identifiers in the sample sheet. ~~Complexity filtering~~ Filtering with bbduk (Bushnell 2014) or PRINSEQ++ (Cantu et al. 2019) removes low-complexity sequences~~containing repetitive elements where~~, while host reads are removed with Kraken2 (Wood et al. 2019).

Metagenomic diversity assessment

Taxonomic classification of preprocessed reads is performed using two complementary approaches - Kaiju (Menzel et al. 2016) and Kraken2 (Wood et al. 2019) - to maximise detection sensitivity across diverse viral families. Results from both classifiers are visualised using Krona (Ondov et al. 2011).

~~De novo~~**De novo** assembly and clustering

The assembly workflow ~~implements~~ employs multi-assembler approaches followed by clustering and scaffolding of contigs independent of the original assembler. *De novo* assembly ~~is performed using~~ uses SPAdes (Meleshko et al. 2021) (RNAviral mode), MEGAHIT (Li et al. 2016), and Trinity (Grabherr et al.

2011), capitalizing on distinct ~~algorithmic strengths~~ algorithms to maximise genome recovery across diverse viral families and ~~variable~~ read depths.

Reference identification uses BLASTn (Altschul et al. 1990) against the Reference Viral Database (RVDB) (Goodacre et al. 2018), retaining the top five hits to ~~facilitate identification of~~ identify related genomic segments ~~and appropriate reference sequences for contig scaffoldingand clustering.~~

for scaffolding. Clustering is performed in two sequential stages. First, taxonomic pre-clustering groups contigs based on taxonomic classification using Kraken2 (Wood et al. 2019) and Kaiju (Menzel et al. 2016), with optional filtering to focus on specific taxonomic clades for more targeted analyses. Second, nucleotide similarity clustering within taxonomic groups is performed using CD-HIT-EST (Li and Godzik 2006) (default), VSEARCH (Rognes et al. 2016), MMseqs2 (Steinegger and Söding 2017), vRhyme (Kieft et al. 2022), or Mash (Ondov et al. 2019). All tools are valid options, though performance may vary depending on the dataset (Zielezinski et al. 2025, Steinegger and Söding 2017).

As an optional filtering step of contig clusters ~~, after~~ following assembly and extension, reads can be mapped to all contigs using BWA-MEM2 (Vasimuddin et al. 2019) (default), ~~BWA (Li 2013), or bowtie2~~ or Bowtie2 (Langmead et al. 2019). Clusters are filtered based on the total percentage of reads mapping to all contigs within a cluster, allowing identification and removal of clusters that likely represent assembly artefacts resulting from low read coverage.

For the final scaffolding step, all cluster members are mapped to the cluster representative or centroid using Minimap2 (Li 2018), followed by consensus calling with iVar (Grubaugh et al. 2019) to generate reference-assisted assemblies. ~~Regions with zero coverage depth can optionally be represented by the reference genome to produce a more complete scaffold genome for consensus calling.~~

Iterative consensus refinement and variant calling

The consensus module supports external reference-based analysis and scaffold refinement. Users can provide ~~a separate reference genome or reference set for each sample with --mapping_constraints; when a reference~~ specific reference genomes; if a set is provided, Mash (Ondov et al. 2019) selects the most similar ~~can be selected using Mash (Ondov et al. 2019).~~

~~Scaffold refinement performs~~ one. The selected reference genome undergoes a single round of refinement to generate the consensus. In contrast, scaffold refinement employs an iterative approach, performing up to 4 iterative cycles (default 2) ~~. Each iteration maps~~ to polish the consensus sequence. Both processes map reads using BWA-MEM2 (Vasimuddin et al. 2019), ~~BWA (Li 2013), or bowtie2 (Langmead et al. 2019) to the consensus~~or Bowtie2 (Langmead et al. 2019), followed by variant calling with BCFtools (Danecek et al. 2021) or iVar (Grubaugh et al. 2019). Benchmarking ~~by Bassano et al. (Bassano et al. 2022) showed that BCFtools outperformed~~ (Bassano et al. 2022) indicated that BCFtools outperforms iVar in precision and recall, ~~where~~ while iVar identified more ~~low frequency~~ low-frequency variants. Users ~~are recommended to consider prioritising~~ can prioritise sensitivity or specificity when selecting the variant caller. Final variant calling is performed on the polished consensus to characterise single nucleotide variants.

Consensus Quality Control

Quality control employs CheckV (Nayfach et al. 2021) ~~for completenessestimates~~to assess completeness, BLASTn (Altschul et al. 1990) for reference similarity, and MMseqs2 (Steinegger and Söding 2017) against the annotated database Virosaurus (Gleizes et al. 2020). These analyses enable species identification, genomic segment classification, host prediction, and any other additional metadata embedded within the reference databases.

The refinement progression is evaluated through sequence alignment with MAFFT (Katoh et al. 2002), which compares final consensus genomes against *de novo* contigs, intermediate consensus sequences from iterative cycles, and the scaffolding reference. All tool metrics are compiled into an interactive MultiQC report (Ewels et al. 2016). Additionally, key metrics are extracted from the MultiQC report and compiled into standalone overview tables to facilitate downstream analysis across all processed samples.

Applications

~~To assess the performance of nf-core/viralmetagenome under challenging scenarios, we simulated coinfection scenarios by mixing paired-end reads from public HIV-1 genomes with varying diversity (80-99%), resulting in 13 samples (See supplementary table 2). Nf-core/viralmetagenome successfully identified coinfections in all mixed samples when genetic similarity was low to moderate ($\leq$ 96.7%).~~

~~We validated nf-core/viralmetagenome performance on~~ We validated the pipeline's performance on a diverse set of real metagenomic samples ~~spanning~~ containing human and plant pathogens. ~~Here, the pipeline successfully generated~~ nf-core/viralmetagenome successfully reconstructed high-quality or near-complete genomes across ~~viral families including segmented~~ various viral families, including segmented viruses (Lassa virus, Orthonairovirus, Tomato spotted wilt tospovirus) and non-segmented viruses (SARS-CoV-2, West Nile virus, Potato virus Y, Youcai mosaic virus, and Monkeypox virus~~).~~

~~Processing~~ ; Supplementary methods S1). The pipeline proved efficient, processing 28 samples ~~(supplementary methods) required~~ in 412 CPU hours ~~and a maximum~~ with a maximum memory footprint of 79GB ~~RAM on an HPC, excluding taxonomic classification steps. The~~ on a high-performance cluster. To facilitate local deployment, we implemented a `-profile local` option, which utilizes a smaller viral RefSeq database instead of the full Kaiju database, reducing peak RAM usage to approximately 18GB.

To evaluate the pipeline's robustness in complex scenarios, we conducted two targeted experiments. First, we assessed its ability to resolve co-infections using a simulated mixture of viral genomes (Supplementary Methods S2). The pipeline successfully distinguished and reconstructed distinct coinfecting strains, provided the genetic divergence was sufficient ($\leq$ 88.7% ANI; Supplementary Figure 1). This capability is crucial for analyzing samples with multiple viral agents or quasispecies.

Second, we investigated the critical role of automated reference selection ~~offers substantial improvements over manual curation by reducing processing time while preserving reconstruction accuracy. Performance correlates strongly with reference database comprehensiveness, as consensus genomes tended to be more complete and similar to the true consensus~~

~~sequence when the scaffolding reference was closer to the true viral genome . This emphasizes the~~ in scaffolding, particularly for segmented viruses with uneven coverage. Using real Lassa virus sequencing data (Supplementary Methods S3), we compared consensus genomes generated using a range of scaffolding references (64% to 100% ANI) against a baseline reconstruction derived from the comprehensive unclustered-RVDB. Our analysis revealed that sequencing depth significantly modulates the reliance on reference quality (Figure 2). For the high-coverage S segment, *de novo* assembly was robust enough to reconstruct the full genome independently, rendering the choice of scaffolding reference largely redundant (Figure 2A-B). In contrast, for the lower-coverage L segment, where *de novo* assembly produced fragmented contigs, the pipeline's ability to identify and utilize a closely related external reference was decisive. We observed a strong positive correlation between the ANI of the scaffolding reference and the completeness and accuracy of the final consensus (Figure 2C-D). This demonstrates that while *de novo* assembly suffices for high-depth regions, the pipeline's automated scaffolding is essential for bridging gaps and recovering complete genomes in low-coverage scenarios.

These findings emphasise the need to keep databases like RVDB (Goodacre et al. 2018) ~~and Virosaurus (Gleizes et al. 2020)~~ up-to-date. Since nf-core/viralmetagenome is primarily designed for eukaryotic viruses, ~~bacteriophage analysis requires different approaches and users~~ users interested in bacteriophage analysis are encouraged to explore pipelines ~~targeting~~ designed to target phages such as VIRify (Rangel-Pineros et al. 2022), VIBRANT (Kieft et al. 2020), VirSorter2 (Guo et al. 2021).

Conclusion

nf-core/viralmetagenome addresses a critical need in viral genomics by providing an automated, scalable solution for untargeted viral genome reconstruction. The pipeline successfully automates the traditionally time-consuming and manual execution process of viral genome ~~assembly~~ reconstruction from short-read metagenomic data through its integrated workflow of *de novo* contig assembly, automated reference selection, clustering algorithms, and iterative refinement strategies.

~~Our validation demonstrates the pipeline 's broad applicability across diverse~~ Our validation results indicate that the pipeline is suitable for a wide range of eukaryotic viral families, ~~achieving~~ it removes the burden of extensive manual curation, and enables consistent, high-quality genome reconstruction ~~while ensuring reproducibility and ease of deployment across different computational environments.~~

with reproducible results across varied computational settings. As viral surveillance and outbreak response increasingly rely on metagenomic sequencing, automated pipelines like nf-core/viralmetagenome will be essential for the timely identification of pathogen strains. The pipeline represents a significant step forward in making viral genome reconstruction accessible to researchers without requiring extensive bioinformatics expertise, facilitating broader adoption of metagenomic approaches in viral research and public health applications.

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Author Contributions

J.K. designed and implemented the pipeline, performed validation analyses, and wrote the manuscript. P.L. and L. E. K. supervised the project and provided critical feedback. The nf-core community contributed to maintaining the pipeline. All authors reviewed and approved the final manuscript.

Conflict of Interest

The authors declare no competing interests.

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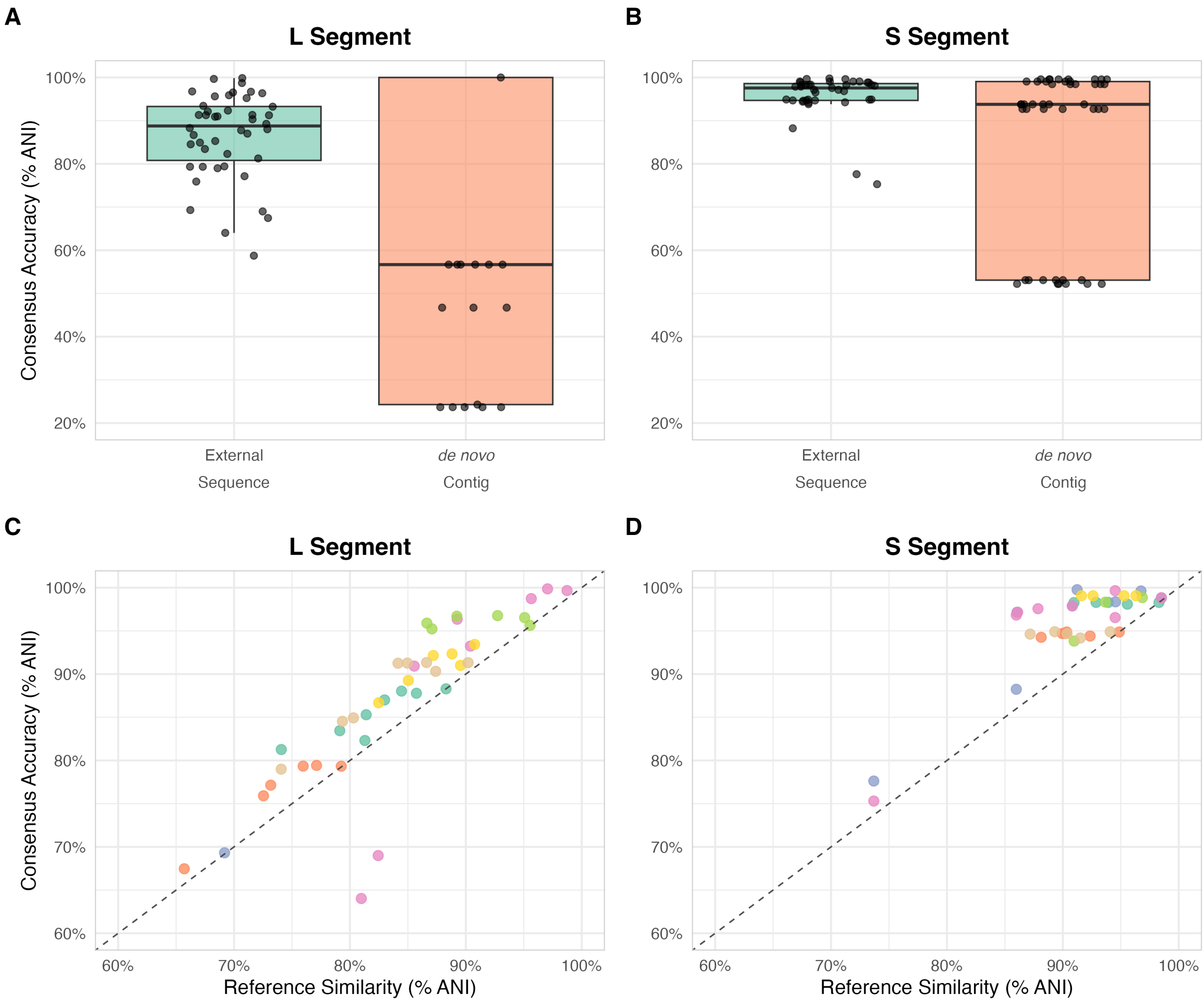

Fig. 2. Impact of scaffolding reference on consensus reconstruction relative to a U-RVDB baseline. (A, B) Consensus genome ANI (%) to baseline using external reference sequences vs. de novo contigs for L and S segments. (C, D) Correlation between scaffolding reference ANI and resulting consensus ANI to baseline for L and S segments, colored by sample.

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